

THE INBOX AGENT

A bilingual inbox assistant for a Pristina electronics shop

A runnable, local prototype for the five challenge cases.

The application uses deterministic phrase matching and business rules, fictional order and policy fixtures, and a bilingual staff interface. It is a demo: it is not connected to a live store, email, Instagram, Viber, or customer database.

1. Agent map

STEP	WHAT THE PROTOTYPE DOES
1	Receive message POST /api/messages accepts message text and optional request metadata.
2	Language, intent and entities Keyword/regex extraction identifies Albanian or English, one of the supported intents, and explicit facts; ambiguous or absent facts remain unknown.
3	Look up mock data Recorded tools query the order and delivery/return/financing policy fixtures. Financing is unconfigured.
4	Apply deterministic rules The rules select AUTO, VERIFY, HUMAN or UNKNOWN. A relationship claim or order number does not verify identity; address details are withheld.
5	Reply and record Template replies are returned in the detected language. HUMAN creates an escalation record; each request gets a sanitized audit record. Tool-call evidence is returned to the interface.

Autonomy and handoffs

AUTO answers within the supported rules and data. VERIFY withholds private details until an approved identity check exists. HUMAN creates an open staff escalation for cases requiring review, including an out-of-policy return or a broken product complaint. UNKNOWN is used when the intent or supported information is insufficient; the reply asks the customer to contact the shop directly and does not claim that a handoff occurred.

Challenge policy facts used

Delivery: 2–4 working days. Returns: within 30 days and unopened. Financing terms are unspecified in the challenge and are not invented by the prototype. All order records are fictional fixtures.

2. Test run: cases 1 and 2

The following are actual outputs from the packaged project's FastAPI application and its included verification setup.

CASE 01 · Delivery delay

INPUT	Porosia #1048 ende s'ka ardhur. Kanë kaluar 6 ditë.
LANGUAGE / INTENT	Albanian (sq) / delivery_status
EXTRACTED FACTS	Order #1048; customer-reported elapsed time: 6 days (kept distinct from working days).
DECISION	AUTO : Order and delivery policy fixtures were found; the configured rule returns AUTO and does not create an escalation for this case.
TOOL EVIDENCE	lookup_order(order_id=1048) → found=True, status=delayed lookup_policy(delivery) → 2–4 working days
ESCALATION	No escalation record created.
CUSTOMER REPLY	Porosia #1048 është shënuar si e vonuar në sistemin tonë. Ju keni raportuar 6 ditë të kaluara që nga porosia. Afati ynë standard i dërgesës është 2-4 working days.

CASE 02 · Return request

INPUT	Can I return headphones after 45 days? Box is open.
LANGUAGE / INTENT	English (en) / return_request
EXTRACTED FACTS	Product: headphones; 45 days since purchase; package opened: yes.
DECISION	HUMAN : The request is outside the standard return conditions; the prototype routes it to a person and promises no refund or exception.
TOOL EVIDENCE	lookup_policy(returns) → within 30 days, unopened
ESCALATION	Created, status open; decision HUMAN.
CUSTOMER REPLY	Your return request for the headphones after 45 day(s) (box opened) falls outside our standard return policy (30 days, unopened only). We can't promise an exception or refund automatically : this has been forwarded to a team member for review.

2. Test run: cases 3 and 4

CASE 03 · Damaged product / repeat complaint

INPUT	3rd time writing! Laptop broken, NOBODY answers!!
LANGUAGE / INTENT	English (en) / damaged_product
EXTRACTED FACTS	Product: laptop; contact attempts: 3; unanswered: yes; broken: yes.
DECISION	HUMAN : A broken product plus repeated unanswered contact is routed for urgent human review.
TOOL EVIDENCE	No tool call.
ESCALATION	Created, status open; decision HUMAN.
CUSTOMER REPLY	We're sorry about the ongoing issue with your laptop. This has been escalated to a team member for urgent review; it has not been resolved automatically.

CASE 04 · Address privacy

INPUT	Arben's brother here. What's the address on order #1031?
LANGUAGE / INTENT	English (en) / order_information
EXTRACTED FACTS	Order #1031; relationship claim: brother. This is not identity verification.
DECISION	VERIFY : The sender is unverified and requests private address information; the data is withheld pending identity verification.
TOOL EVIDENCE	lookup_order(order_id=1031) → found=True, status=not disclosed verify_customer → verified=False; private details withheld
ESCALATION	No human escalation record; identity verification is required before private data can be shared.
CUSTOMER REPLY	For security reasons, we can't share private details of order #1031 (such as the delivery address) without first verifying your identity through an approved method. Please reach out so we can start verification.

2. Test run: case 5

CASE 05 · Installment question

INPUT	A mund ta blej laptopin me këste?
LANGUAGE / INTENT	Albanian (sq) / installment_purchase
EXTRACTED FACTS	Product: laptop; financing terms are not configured in the demo data.
DECISION	UNKNOWN : No financing policy is configured, so terms are not guessed and the outcome remains UNKNOWN.
TOOL EVIDENCE	lookup_policy(financing) → not configured / unknown
ESCALATION	No escalation record created.
CUSTOMER REPLY	Aktualisht nuk kemi një politikë të konfirmuar financimi/me këste për laptop, kështu që nuk mund t'i përgjigjemi kësaj në mënyrë të sigurt me informacionin që kemi. Ju lutemi kontaktoni dyqanin drejtpërdrejt për opsionet e financimit.

Verification summary

The packaged verification scripts were run on Windows. The checkpoint 6 end-to-end checks passed, covering /health, all five challenge messages, decisions, tool evidence, escalation listing, audit lookup, and protection against private address disclosure. The checkpoint 7 reply-wording checks also passed for UNKNOWN and HUMAN outcomes.

3. One failure and fix

Failure found during testing

A prior reply template used wording that implied an UNKNOWN case had been forwarded to staff, even though the pipeline created no escalation record. This could mislead a customer into expecting a follow-up that had not been arranged.

Fix

Reply templates were changed so a team handoff is stated only when the decision is HUMAN and an escalation is created. UNKNOWN replies now explain that the system cannot safely answer from available information and direct the customer to contact the shop.

Retest

The included verify_checkpoint7.py checks UNKNOWN installment, mixed-intent and unknown-order cases for the absence of handoff claims, and confirms that the HUMAN return and damaged-product cases still state the handoff. It passed when run against this project.

4. Business case

At 250 messages per day over five working days, the inbox receives 1,250 messages per week; if an assumed 60% (750 messages) can be answered autonomously and each would otherwise take an assumed four minutes, estimated time saved is $750 \times 4 \div 60 = 50$ hours per week.

The 60% automation rate and four-minute handling time are planning assumptions, not measured results, and the five challenge cases do not measure the shop's actual inbox mix or automation rate.

The biggest risk is exposing private order details or making unsupported promises; mitigation is to require verification before private-data disclosure, apply deterministic policy checks, use UNKNOWN when evidence is missing, escalate exceptions to a person, and retain sanitized audit records.